Preliminary Sample-Size Analysis for CLAPS

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This note reports preliminary sample-size guidance for CLAPS (A Crosslinguistic Approach to Passive Semantics). CLAPS studies how the acceptability of passive and related sentence types depends on the semantic affectedness of the patient, and how that dependence differs across sentence types and languages. The guidance comes from a Bayes-factor design analysis running on the University of Oxford Advanced Research Computing cluster. A design analysis of this kind simulates the planned model many times under plausible true effects, then asks how often a given sample size would yield compelling evidence for the focal predictions.

Two versions of the analysis are run. The primary one is data-grounded. It fits the planned model to each language's real pilot data and simulates new studies from that fit, so it conditions on the effects actually observed in the CLAPS pilot, which is the basis the collaborators asked the design analysis to use. A single pilot's point estimate is noisy and tends to overstate power, the concern that Albers and Lakens (2018) raise for pilot-based power analyses. The data-grounded analysis avoids that point estimate in two ways. First, it draws the focal effects from the pilot posterior on each simulated study, so that the reported probability of a decisive Bayes factor integrates the uncertainty in the estimated effect. This yields an assurance rather than a point power. Second, it also evaluates a deliberately conservative lower bound on the effect, as a safeguard against an optimistic pilot. A second version, for comparison, conditions instead on the pooled effect sizes from the five previously published studies. It provides an independent cross-check against the broader evidence base.

The two versions agree on the central structural point. The power to detect the affectedness effect is governed by the number of verbs in the materials rather than by the number of participants, because affectedness is a property of the verb. They diverge on whether a modest sample is enough. On the published-literature effects, a per-language sample of at most 50 participants would suffice at the full verb set. On the pilot effects, and with the verb-level variability measured from the pilot rather than assumed, the picture is less favourable. Norwegian reaches the target at about 100 participants, but English and Turkish do not reach it at any simulated sample size. For those two languages, the binding constraint is the number of verbs, and in Turkish also the strength of the effect itself, so adding participants alone cannot bring the design to the usual power target. Because the data-grounded analysis is the agreed primary basis, this is the finding that should guide the design. The report quantifies these limits analytically, explains why the two analyses disagree, and sets out design options that remain feasible for a volunteer consortium. The recommended option, a pooled cross-linguistic analysis, is now built, validated and running on the cluster, and the section on it below gives its rationale and the results to expect.

The data-grounded analysis is substantially complete but still running. Results are in for all three languages up to 130 participants, for English up to 400 and for Turkish up to 400, and the largest samples are still computing. Each simulated condition rests on 22 to 40 studies, so the individual figures carry a Monte-Carlo error of several percentage points. The English plateau, which the 400-participant cells now confirm, no longer rests on extrapolation. The report should be read as a decision-relevant interim pass rather than a final recommendation.

Background and Motivation

The question CLAPS addresses is whether the acceptability of passive and related sentence types tracks the semantic affectedness of the patient, and whether that relationship differs by sentence type. Affectedness, the degree to which a patient or theme undergoes a change of state or is causally impinged upon by the action, is a core dimension of argument structure (Beavers, 2011; Dowty, 1991). It has long been argued to be the semantic heart of the passive, which carries the meaning that the surface subject is in a state or circumstance characterised by the agent having acted upon it (Pinker et al., 1987). Across languages, adult acceptability-judgement and comprehension studies support the view that passive acceptability is semantically structured rather than purely syntactic (Ambridge et al., 2016; Ambridge et al., 2023; Aryawibawa & Ambridge, 2018; Bidgood et al., 2020; Darmasetiyawan & Ambridge, 2022; Liu & Ambridge, 2021).

The cross-linguistic evidence base to date spans English, Indonesian, Balinese, Hebrew and Mandarin. A Bayesian meta-analytic synthesis finds a consistent pattern across them (Ambridge et al., 2023). Affectedness raises acceptability for both passives and actives, more strongly for passives, and does not raise it for pseudo-passives. The affectedness constraint has not yet been tested in Turkish or Norwegian. CLAPS extends the paradigm to these two typologically and morphologically distinct languages. No prior language-specific effect-size estimates exist for them, so the CLAPS pilot supplies the first such estimates. The data-grounded analysis uses these pilot estimates as its primary basis, and the cross-linguistic literature provides an independent comparison, which turns out to be the more optimistic of the two.

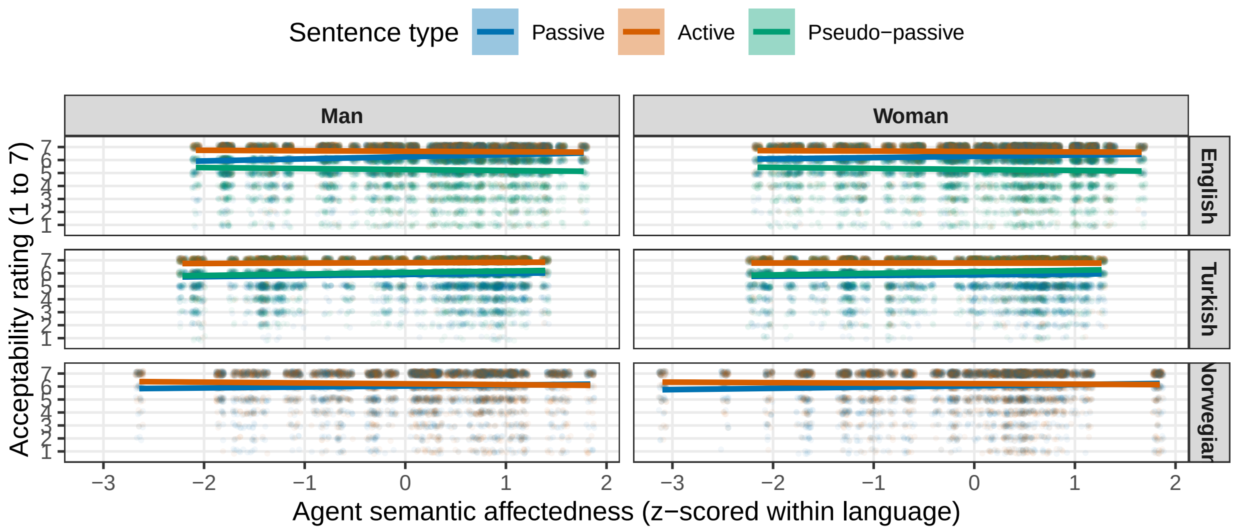
The Acceptability Model and the Focal Predictions

Each acceptability judgement is a discrete rating from 1 to 7. These ratings are modelled with a Bayesian cumulative-logit mixed-effects model, an ordinal model that respects the discreteness, ordering and bounded range of rating data while leaving the underlying scale free to vary across conditions (Bürkner & Vuorre, 2019). Treating the same ratings as continuous, under a Gaussian likelihood, biases the regression coefficients and inflates error rates where floor or ceiling effects are present (Liddell & Kruschke, 2018), and recent empirical work shows that modelling them cumulatively recovers psycholinguistic effects more accurately (Taylor et al., 2023; Veríssimo, 2021).

The expected pattern is already visible in the pilot data. Figure 1 shows the raw acceptability ratings against agent semantic affectedness, separately by sentence type, referent gender and language. The affectedness slope is positive and steepest for the passive in English and Norwegian, and shallower for the active and the pseudo-passive. The trends are very similar for male and female referents within each language. This is reassuring, because the focal effects should not be driven by a referent-gender imbalance.

Figure 1

Raw Pilot Acceptability by Agent Semantic Affectedness, Sentence Type, Referent Gender and Language



Note. Points are individual pilot ratings, jittered. Lines are per-sentence-type linear trends with 95% confidence bands. The pilot data are used for calibration only. Norwegian has no pseudo-passive.

The model predicts acceptability from sentence type, with the passive as the reference level and the active and the pseudo-passive as comparisons, from verb affectedness, and from their interaction. The focal predictions concern the affectedness slope in the passive and how it changes across sentence types. The primary prediction (H1b) is one-tailed and holds that affectedness raises passive acceptability more than it raises active acceptability, so the active-by-affectedness interaction is negative. A supporting prediction (H1a) holds that affectedness raises acceptability in the passive itself, so the passive affectedness slope is positive. A sample size is judged adequate only when both of these predictions reach the evidence threshold. A secondary, two-tailed prediction (H2) concerns the pseudo-passive-by-affectedness interaction. It does not gate the recommendation and does not apply to Norwegian, which has no pseudo-passive.

Design-Analysis Method

The design analysis estimates statistical power by simulation. Each simulated dataset is drawn from the same cumulative-logit model that will be used for analysis, under assumed values for the effects of interest. Each dataset is then analysed in the same way. The proportion of simulated studies that yield decisive evidence estimates the power at that sample size. The whole procedure stays within the Bayesian framework. This matters because the confirmatory test is a Bayes factor rather than a significance test, and the power to return a decisive Bayes factor is not the same quantity as the power to return a significant p-value.

A central feature of the design is that affectedness is a property of the verb rather than of the individual trial. Each verb carries a single affectedness value, held constant across participants and sentence types, exactly as in the real materials, where every verb has one affectedness score. The number of verbs, rather than the number of trials, therefore determines the precision with which the affectedness effect can be estimated. The verb count is treated as a design dimension alongside the participant count.

The two analyses differ in where they take the assumed true effects and the affectedness values. The data-grounded analysis, which is primary, fits the planned model to each language's pilot data. It then takes the population effects, the random-effect covariance and the ordinal thresholds from that fit, and it reads each verb's actual affectedness score from the pilot data. The literature-anchored cross-check instead sets the population effects to values from the cross-linguistic meta-analysis. On the standardised affectedness predictor, the passive affectedness slope is set near the pooled value of about 0.23 logits per standard deviation, with a negative active-by-affectedness interaction of about -0.15 and a small positive pseudo-passive-by-affectedness interaction. In the cross-check, affectedness is spread across a plausible range rather than read from the pilot, and the by-participant and by-verb random-effect standard deviations are set to realistic magnitudes. The referent-gender covariate is held out of both preliminary runs and is examined separately.

The analysis model is the maximal random-effects structure consistent with the design. It has correlated by-participant slopes for the full sentence-type-by-affectedness structure, together with by-verb intercepts and sentence-type slopes (Barr et al., 2013). Because affectedness does not vary within a verb, a by-verb affectedness slope is not identifiable and is omitted, so affectedness enters as a population-level effect. The priors are weakly to moderately informative and centred at zero on the focal slopes, so that the directional tests are not pre-loaded. A normal prior of standard deviation 0.5 is placed on the affectedness slope and on the active interaction, and a slightly wider normal prior of 0.6 on the pseudo-passive interaction. The sentence-type main effects take a wider normal prior. Student-t priors are placed on the ordinal thresholds and on the group-level standard deviations, and an LKJ prior on the group-level correlations (Gelman, 2006; Lewandowski et al., 2009).

Evidence for each prediction is summarised by a directional Savage-Dickey Bayes factor, obtained by sampling the prior alongside the posterior (Wagenmakers et al., 2010). A simulated study counts as a detection when its Bayes factor reaches 10, the preregistered threshold for strong evidence. Power is the proportion of simulated studies that reach it. The recommended sample size is the smallest at which both focal predictions reach a power of 80%. Each replicate is fitted with four Markov chains of 3,000 iterations after 1,000 warm-up iterations. The literature-anchored grid reported here crosses the three languages with participant counts from 50 to 120. The cells available so far reach 90 for English and 60 for Turkish at the design's verb counts, with verb counts of 40 and 72, close to the design's actual count of 66 to 72. Each cell holds 50 replicates. This gives a Monte-Carlo standard error of about five to six percentage points near a power of 80%, so estimates close to the threshold are not perfectly ordered across adjacent sample sizes. The central code for the data-generating process, the model and the Bayes factor is reproduced in Appendix B.

Results

Data-grounded results

The data-grounded analysis is the primary basis for the recommendation, and it is now substantially complete. In its assurance form it draws the focal effects afresh from each language's pilot posterior on every simulated study, so the reported power carries the pilot's own uncertainty about the size of those effects. Table 1 reports the joint power, the proportion of simulated studies in which both focal predictions clear the Bayes-factor threshold together, at the pilot's real verb count.

Table 1

Preliminary Data-Grounded Joint Power by Participant Count and Language, Assurance Variant

Participants	English	Turkish	Norwegian
30	25%	21%	42%
50	42%	38%	67%
70	38%	33%	75%
100	54%	25%	88%
130	46%	29%	88%
150	62%	35%	-
200	65%	42%	-
250	70%	42%	-
300	72%	36%	-
400	62%	41%	-

Note. Joint power is the proportion of studies in which both focal predictions clear a Bayes factor of 10 together, at 72 verbs and 66 for Norwegian. Cells below 10 replicates are still accumulating and shown blank. The rest hold 22 to 40 studies each.

The pattern differs sharply by language. Norwegian reaches the 80% target at about 100 participants and is comfortably powered beyond it. English and Turkish do not reach it at any simulated sample size. For English, the joint power rises with sample size only slowly and settles below the target, at 72% by 300 participants and 62% at 400. The reason is structural. The joint requirement is held back by the affectedness main effect (H1a), whose detection rate levels off at roughly three in four, while the interaction (H1b) climbs comfortably past it. Because affectedness is a property of the verb, its main effect is estimated from the 72 verbs, so adding participants does not sharpen it. The precision of that effect, and with it the power to detect it, is set by the number of verbs. For Turkish, both focal effects are weak in the pilot. The joint power never exceeds 42% at any simulated sample size, and the present design does not bring it to target.

A conservative safeguard variant, which fixes each focal effect at a cautious lower bound of its pilot posterior, is lower again. Its highest joint power over the samples run is 58% for Norwegian, 25% for English and 12% for Turkish. This is the design's power under a deliberately pessimistic reading of the pilot.

Computed directly from the pilot posterior, the probability that both focal effects have their predicted sign is 99.6% for English, 99.5% for Norwegian and 94.0% for Turkish, so the pilot points the right way in all three languages. The shortfall for English and Turkish is therefore a matter of precision rather than direction. Because affectedness varies only between verbs, the standard error of its effect cannot fall below the by-verb variability divided by the spread of the affectedness scores, however many participants respond. This is the stimulus-sampling limit long recognised in psycholinguistics (Clark, 1973) and quantified for designs with crossed random effects by Westfall et al. (2014; see also Judd et al., 2017; Brysbaert & Stevens, 2018). The limit can be computed directly from the fitted pilot models, with no simulation involved. Table 2 reports the resulting upper bounds, the detection rates that would remain even with unlimited participants at the current verb counts.

Table 2

Analytic Upper Bounds on Detection with Unlimited Participants, at the Current Verb Counts

Language	H1a ceiling	H1b ceiling	Joint ceiling	Verbs for 80%	Verbs for 90%
English	83%	97%	81%	72	124
Turkish	77%	65%	49%	348	> 1,200
Norwegian	85%	100%	85%	66	98

Note. Ceilings are the probability of a Bayes factor of at least 10 with unlimited participants, from the 8,000 pilot posterior draws and the fitted by-verb covariance at 72 verbs and 66 for Norwegian. The last columns give the verbs needed for an 80% and 90% joint ceiling. A ceiling just above target, as for English, still leaves it out of reach, since any finite sample sits below its ceiling.

These bounds agree closely with the simulations. English's simulated joint power climbs to 72% by 300 participants against a ceiling of 81%. Turkish never exceeds 42% against a ceiling of 49%. Norwegian's simulated 88% at 100 participants matches its ceiling of 85% within Monte-Carlo error. The agreement between two entirely different routes to the same quantities, stochastic simulation and direct calculation, is a strong check that the pipeline is sound. A further check makes the point starkly for English. The pilot's own posterior standard deviation for the affectedness slope, 0.19, already sits close to the verb-count floor of 0.19. A replication with the same 72 verbs would therefore have little room to estimate that slope more precisely than the pilot already has, whatever its sample size. The residual gap reflects the floor formula's own approximation, since it ignores some of the covariance in the by-verb random effects, rather than a real target that more participants could close.

The ceilings also show what would and would not help. Relaxing the evidence criterion from a Bayes factor of 10 to 6 or 3 lifts the English joint ceiling to 86% and 92%, but lifts Turkish only to 58% and 70%, so a weaker threshold cannot rescue the Turkish test. Adding verbs raises every ceiling, because the precision floor shrinks with the square root of the verb count. English would reach a joint ceiling of 90% at about 124 verbs, and Norwegian at about 98. Turkish would need about 348 verbs merely to reach a ceiling of 80%, which is not realistic within a single language, so for Turkish the remedy lies elsewhere, as taken up under Paths Forward below.

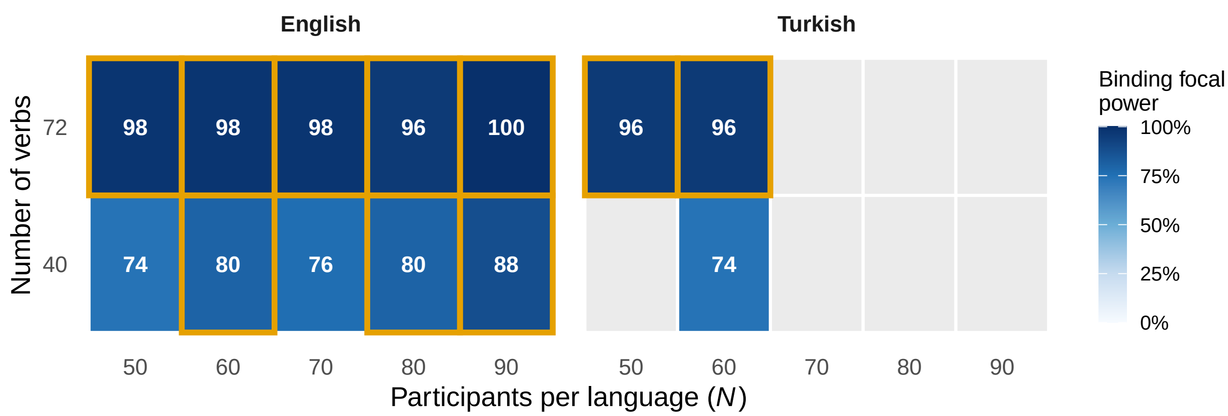
These results are preliminary in one remaining respect, since each condition rests on 20 to 40 simulated studies and so carries a Monte-Carlo error of several percentage points. The English plateau itself, though, is now confirmed rather than extrapolated. At 400 participants, with 32 replicates in, the joint power sits at 62%, flat against the 300-participant value rather than rising towards the target. Overturning the plateau would have required a sustained rise at 400 and 500, and the 400-participant data already rule that out. The largest cells still computing will complete the picture, but on that criterion they can no longer reverse it.

Literature-anchored cross-check

The grid is read as a surface over participant count and verb count, summarised by the binding focal power, the smaller of the two focal detection probabilities, since the recommendation requires both to clear 80%. This cross-check is the more optimistic of the two analyses, because it fixes the effects at the pooled literature values and carries no uncertainty about them. Figure 2 shows this surface for the two languages whose cells are complete. Cells outlined in gold meet the target, and blank cells are those whose replicates are still accumulating. The pattern is clear. Power increases steeply with the number of verbs and only weakly with the number of participants.

Figure 2

Binding Focal Power by Participant Count and Verb Count, for Each Language



Note. Each cell gives the binding focal power as a percentage, with darker shading for higher power. Cells outlined in gold meet the 80% target. Blank cells are still accumulating.

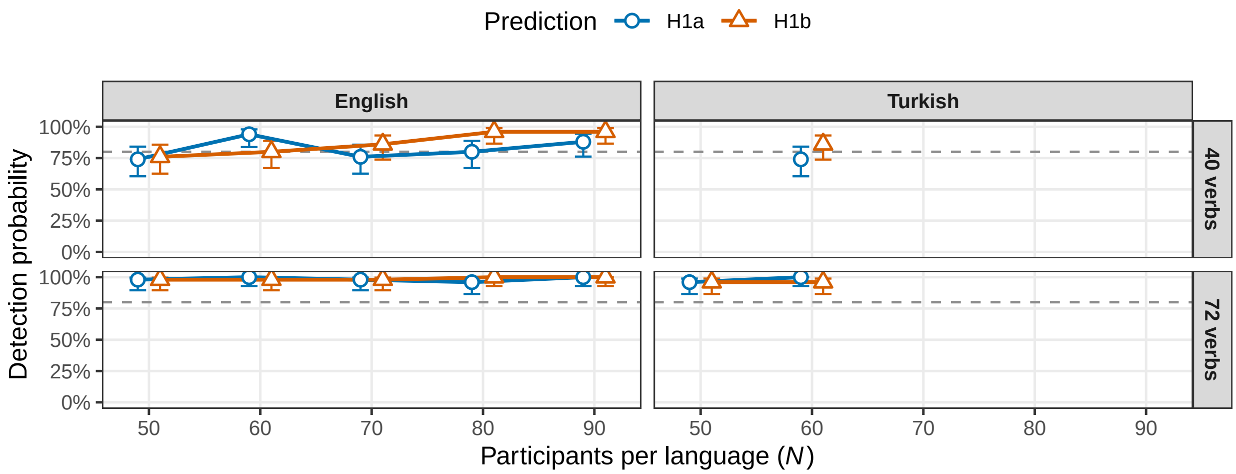
The two axes are not equally important. At 40 verbs in English, the binding power is 74% at 50 participants and reaches the 80% target only between about 60 and 90 participants, within Monte-Carlo noise. At 72 verbs, it is already 96% to 98% at 50 participants. Increasing the number of verbs from 40 to 72 raises power more than increasing the number of participants does, which follows from affectedness being a verb-level property. Figure 3 separates the two focal predictions and shows how each approaches the target as the sample grows, at each verb count.

Note. Points are detection probabilities over the simulated studies in each complete cell, with Wilson 95% intervals reflecting Monte-Carlo uncertainty. The dashed line marks the 80% target. Points are dodged so the two predictions do not overlap.

At the design's real verb count, which is 72 for English and Turkish and 66 for Norwegian, the binding focal power on these literature effects is already between 96% and 98% at 50 participants, the smallest sample simulated, in both languages observed so far. On the literature effects, the adequate per-language sample is therefore at most 50 participants at the full verb set. This is the optimistic reading. The data-grounded results above, which condition on the weaker pilot effects, do not reach the target for English or Turkish at any sample size. At 40 verbs, the design is borderline, reaching the target only between about 60 and 90 participants within Monte-Carlo noise. Because 50 is the smallest sample in the main grid, the analysis places the adequate sample at or below 50 without determining it more precisely. A dedicated low-N arm, at participant counts of 20, 30 and 40 with the full verb set, is now running to estimate the smallest adequate sample, and the result will be added here. Norwegian is expected to follow the same pattern and is still computing.

Figure 3

Detection Probability for the Focal Predictions, with Wilson 95% Intervals, by Verb Count and Language



Why the Two Analyses Disagree

The gap between the two analyses calls for an explanation, because the literature-anchored grid reports 96% to 98% at 50 participants on the same 72 verbs at which the data-grounded joint power never reaches 80%. Expressed per standard deviation of affectedness, the pilot's passive slope is 0.26 logits for English, 0.23 for Norwegian and 0.18 for Turkish, against a pooled literature value of about 0.23, and the Turkish active interaction, at −0.15, matches the literature's −0.15 almost exactly. The English interaction, at −0.44, is almost three times the literature value, so the CLAPS pilot effects are not unusually small.

The disagreement comes instead from the noise side of the simulation, in two specific assumptions. First, the literature-anchored simulator assumed a by-verb intercept standard deviation of 0.5, where the fitted pilot models put it at 0.83 for English, 0.64 for Turkish and 0.66 for Norwegian. The precision floor on the affectedness slope scales directly with this quantity, so assuming 0.5 where the fitted value is 0.83 overstates the attainable evidence considerably. Second, the literature-anchored simulator included no by-verb sentence-type slopes at all, where the fitted pilots put their standard deviation at 0.88 for English, 0.64 for Turkish and 0.34 for Norwegian. That omission left the interaction test with no verb-level noise floor in the simulation, which is why the earlier grid found the interaction so easy to detect. Both assumptions were reasonable defaults before the pilot models existed. The pilot has now measured the quantities they stood in for, and the data-grounded analysis uses the measured values. The earlier optimism, in short, came less from the assumed effect sizes than from understating how much verbs vary (Clark, 1973; Westfall et al., 2014).

Effect-Size Qualification

One assumption deserves particular care, because it bears directly on the absolute sample size. The numbers shown here condition on assumed effects set to the cross-linguistic pooled posterior means from a Bayesian meta-analytic synthesis of the passive-affectedness literature, which is a central estimate rather than an optimistic one (Ambridge et al., 2023). On the simulator's internal predictor scale, these appear as larger nominal values, but expressed per standard deviation of affectedness, they are about 0.23 and -0.15 logits, matching the pooled literature values. The larger nominal numbers therefore reflect the scaling rather than a larger effect. Two cautions nonetheless apply. The first concerns the noise assumptions rather than the effects. As the previous section sets out, the fitted pilot models put the verb-level variability well above the values this grid assumed, and that difference, more than any difference in effect size, is what made the grid optimistic. The pilot effects themselves are broadly in line with the pool per standard deviation of affectedness, though the Turkish terms sit at the weak end, with the interaction at about -0.15 in those units. The second is that the literature is itself subject to publication bias and to the winner's curse, the tendency of published estimates, especially from small and low-powered studies, to overstate true magnitudes (Button et al., 2013; Gelman & Carlin, 2014; Ioannidis, 2008). Design analyses anchored to such estimates can therefore be optimistic about power (Albers & Lakens, 2018; Vasishth et al., 2018).

These cautions are the reason the data-grounded analysis, rather than this cross-check, is the primary basis for the recommendation. It draws the focal effects from each language's own pilot posterior, and it also evaluates a conservative lower bound. It therefore conditions on the effects observed in CLAPS and carries their uncertainty through to the reported power. Its results are reported above. They agree with this cross-check for Norwegian, which is powered at about 100 participants, and they diverge from it for English and Turkish, where the weaker pilot effects hold the joint power below target at every sample size run.

A separate point concerns the scale on which the prior sits, rather than the size of the assumed effect. The literature-anchored grid behind the figures expresses affectedness on the simulator's internal predictor, whose standard deviation is about 0.29. The real confirmatory analysis and the data-grounded analysis both place it on the analysis pipeline's Gelman scaling, whose standard deviation is 0.5, while the same weakly informative prior is applied throughout. A Bayes factor reflects the width of the prior relative to the coefficient it constrains. Holding the prior fixed while the predictor scale differs therefore leaves the literature-anchored grid mildly optimistic about the focal-slope Bayes factor, by a factor of about 1.7, relative to the analysis that will actually be run. The qualitative conclusion is unaffected, because the verb count dominates the result, the focal Bayes factors at the full verb set lie far above the threshold, and the data-grounded analysis already uses the correct scale. A scale-corrected version of the cross-check grid is being computed to confirm the per-language sample sizes directly. Until it completes, the figures here should be read as a mild upper bound on the Bayes-factor evidence.

For an effect that lives at the level of the participant, a weaker true effect simply needs a larger sample, and the rough guide is that a true effect around two-thirds of the assumed magnitude roughly doubles the sample needed. The affectedness main effect lives at the level of the verb, however, where its precision is set by the number of verbs, so a weaker true effect cannot be recovered by recruiting more participants. This is exactly what the data-grounded results show. At the full verb count and the pilot's own effects, the affectedness power for English settles below the target and does not climb with sample size, and for Turkish neither focal effect is strong enough to bring the joint power to target. The effective levers are therefore the number of verbs, and for Turkish the strength of the interaction, not the participant count. Where feasible, the design should still be planned against a deliberately conservative, or safeguard, effect size rather than a point estimate (Perugini et al., 2014), and the data-grounded analysis builds this in directly through its assurance and safeguard variants. A dedicated sensitivity arm makes the same point for the literature-anchored grid. It re-evaluates the 72-verb design at 75% and 60% of the assumed effect magnitude, with the

60% level approximating the weaker effects seen in the pilot. Its discounted-effect sample sizes will be added once that arm completes.

Convergence and Computation

Convergence is clean across the grid, with no divergent transitions anywhere. At the smaller verb count, essentially every fit meets the strict criteria, namely an R-hat below 1.01 together with bulk and tail effective sample sizes of at least 400. At 72 verbs, the maximal model occasionally falls just short of these criteria. This happens in roughly one fit in ten at the smaller samples, and up to one in five at the largest, where N is 80 to 90. It occurs on the R-hat or effective-sample-size thresholds rather than on divergences, which reflects sampling efficiency rather than a specification failure. Restricting the power calculation to the fully converged fits leaves the estimates essentially unchanged. For English at 72 verbs and 50 participants, the binding focal power is 98% over all 50 fits and 98% over the 46 converged fits, so the recommendation is robust to this. Per-fit runtime grows steeply with the verb count, reaching between one and two hours at 72 verbs.

Secondary Prediction

The pseudo-passive-by-affectedness interaction reaches a detection probability of about 58% to 80% at 72 verbs. This remains below the focal predictions and reflects its smaller assumed magnitude, and the opposite-direction test is near zero throughout, as expected. This prediction is secondary and two-tailed, and it does not constrain the sample-size recommendation. Full operating characteristics for all four predictions, including the convergence rates, are reported in Appendix C.

Paths Forward

The analytic bounds turn the shortfall into a set of concrete design choices. Four options follow, in roughly increasing order of departure from the current plan. Since they were drafted, the collaborators have fixed the design at 80 participants per language and 72 verbs, the most a volunteer team can collect in a session of about 45 minutes. The options are read against that cap, and the recommendation at the end is a two-tier design that draws on several of them.

The first option changes nothing about the materials and revisits the adequacy rule. The design's primary prediction is H1b, the active-by-affectedness interaction, with H1a as a supporting prediction. Gating each language on the primary prediction alone, with the passive slope reported as estimation, leaves English adequately powered from about 130 participants and Norwegian from about 50, at the current verb counts. The joint gate on both predictions is what no feasible English sample can satisfy. This option does not help Turkish, whose interaction ceiling is 65% even on its own.

The second option makes the cross-linguistic claim the primary confirmatory test. The theoretical claim CLAPS addresses is explicitly cross-linguistic, that affectedness structures passive acceptability across languages, and the meta-analytic synthesis of the earlier studies frames the evidence in exactly those terms (Ambridge et al., 2023). A hierarchical model fitted to all three languages jointly tests that claim directly. It also multiplies the verb-level information, roughly 210 verb-by-language units in place of 72, which divides the precision floor by about the square root of three. On the pilot effects, every language's focal terms point the predicted way, so the pooled test starts from a favourable position. Turkish then contributes fully to the confirmatory result, and its language-specific effects are reported as estimates with credible intervals, which reflects honestly what a 72-verb single-language design can support. The pooled design analysis runs on the same machinery used here, and it is now built and on the cluster. Its rationale and expected results are set out below.

The third option expands the verb pool without lengthening any participant's session. Verbs, not participants, are the scarce resource, and a planned-missingness design assigns each participant a random subset of a larger pool (Graham et al., 2006; Little & Rhemtulla, 2013). Each volunteer still rates about 72 items, so the 45-minute session is unchanged, while the pool grows to, say, 120 to 150 verbs. At about 124 verbs the English joint ceiling reaches 90%, and the pooled cross-linguistic test gains in proportion. The cost falls on materials, namely affectedness norms and translations for the new verbs, rather than on participant time. Whether this option is available turns on the agreed 72-verb limit being a constraint on the session, which planned missingness respects, rather than a cap on the total pool. Turkish alone would need about 348 verbs, so verb expansion supports the pooled test rather than substituting for it.

The fourth option spends volunteer capacity more efficiently rather than demanding more of it. A sequential design analyses the data in waves and stops when the Bayes factor reaches a preset bound or a maximum sample is reached, which typically requires markedly fewer participants than a fixed-sample design with the same operating characteristics (Schönbrodt et al., 2017; Stefan et al., 2019), and optional stopping poses no problem for Bayes-factor inference (Rouder, 2014). Sequential monitoring cannot lift the verb-count ceilings, but it minimises the participants needed to reach whatever evidence the design can deliver, and it fits a distributed, volunteer-based collection model of the kind that large multi-lab consortia have made standard (Moshontz et al., 2018).

The recommendation that follows from the numbers is a two-tier design. The pooled cross-linguistic analysis, the second option, carries the confirmatory test at the strong threshold, since that is where the theory lives and where the verb-level information is greatest. Each language then reports moderate evidence at a Bayes factor of 3 and full estimation, which is what a capped single-language design can honestly support. Data collection proceeds in waves with sequential monitoring, the fourth option, so volunteer labs stop once the evidence is in. The verb pool is enlarged under planned missingness, the third option, if the 72-verb limit permits. Turkish contributes fully to the pooled result and is reported as estimation, since no per-language criterion brings it to target. The pooled design analysis sets the per-language target within the 80-participant cap, and the first pooled cells already give strong evidence at 50 participants per language, so the target is likely to sit within reach of a volunteer consortium. A dedicated run at 70, 80 and 100 participants, now on the cluster, will fix the exact figure.

The Pooled Analysis: Rationale and Expectations

Following the recommendation above, the pooled cross-linguistic design analysis has been implemented, validated and submitted to the cluster. Each simulated study draws every language's effects from that language's own pilot posterior, exactly as in the per-language assurance runs, simulates the three datasets, and analyses them together with the cross-language model used in the earlier synthesis work, which carries correlated random slopes by participant, by verb and by language. The evidence criteria are unchanged, namely directional Savage-Dickey Bayes factors on the pooled focal coefficients under the same zero-centred priors. Following the collaborators' cap of 80 participants per language, the pooled grid covers samples from 50 to 100, with thirty simulated studies per condition. A companion job fits the same pooled model to the combined real pilot data, which checks convergence on real data and supplies the pooled analytic ceiling. The first pooled cells have returned. At 50 participants per language, across the 9 replicates available so far, the joint power is 78% at the strong-evidence threshold and 100% at a Bayes factor of 3, with median Bayes factors of 11 and 16 for the affectedness slope and the interaction. These sit in line with the analytic expectation that the pooled test is well powered within the cap, and they remain preliminary while the larger samples compute.

The rationale is theoretical before it is statistical. The hypothesis CLAPS tests is cross-linguistic, that passive acceptability tracks affectedness across languages, and the evidence for it has always been read across languages, most explicitly in the meta-analytic synthesis of the five earlier studies (Ambridge et al., 2023). The per-language analyses treat each language as a self-contained confirmatory study, which is exactly why each inherits its own verb-count ceiling. The pooled model instead tests the general claim directly. Its population-level affectedness slope and interaction are the cross-linguistic effects, and its by-language random slopes let each language depart from them, so generality is tested rather than assumed.

Statistically, pooling attacks the binding constraint identified above. The precision floor on the affectedness effect is set by the verb-level information, and the pooled analysis draws on about 210 verb-by-language units instead of 72, which shrinks the floor by roughly the square root of three. The pilot slopes point the predicted way in all three languages, at 0.52, 0.36 and 0.47 logits for the affectedness main effect and -0.88, -0.29 and -0.53 for the interaction, so their precision-weighted average sits comfortably clear of zero. Partial pooling also shares strength in the other direction, since each language's estimate borrows from the others in proportion to its own uncertainty, and Turkish gains most from that.

The expected differences from the per-language results follow from this. First, the ceilings should effectively disappear as constraints. Setting the pooled effects near the averages above against a floor of about 0.10 logits puts both pooled focal ceilings near 100%. The pooled test should therefore be limited by ordinary sampling noise rather than by a structural cap, and the joint power should cross 80% within the swept range, plausibly between about 70 and 130 participants per language. Second, the limiting quantity changes. What holds the pooled test back is no longer the verb count but the variation between languages, especially in the interaction, whose pilot values span -0.29 to -0.88. That variation does not shrink with more verbs or more participants, and with only three languages its variance is weakly identified and partly prior-driven (Gelman, 2006), so it is the number to watch in the results and in sensitivity checks. Under the pilot values, it still leaves the pooled ceilings near 97% or higher. Third, the interpretation changes. A strong pooled Bayes factor alongside a wide Turkish-specific interval is a coherent and likely outcome, and it is the same pattern the cross-linguistic literature shows. The per-language conclusions above stand unchanged. Norwegian is confirmable on its own, and the English and Turkish per-language joint tests remain capped whatever the pooled result.

If the simulations return materially weaker power than these expectations, the likely cause is the between-language variance component rather than the focal effects, and the pooled test's sensitivity to that component's prior will be examined before any recommendation is drawn from it.

Limitations and Status

The status differs across the two analyses. The literature-anchored cross-check is complete for English and Turkish and points to a modest sample at the full verb set. The data-grounded analysis, which is the primary basis, is substantially complete. Norwegian is finished and is adequately powered at about 100 participants. English and Turkish fall short of the joint power target at every sample size run so far, and only their largest cells are still computing. The finding that participants alone cannot bring English and Turkish to target rests on the affectedness effect being verb-limited, the 400-participant cells now confirm it for English at 32 replicates, and the analytic ceilings bound it from above, so the running cells are expected to confirm the plateau rather than change its direction. One technical note on the assurance draws belongs here. Each replicate reuses one of the first 24 to 40 saved posterior draws rather than a thinned random sample. Checked against the full posterior, those blocks are representative for English and Norwegian and sit slightly below the mean for the Turkish affectedness slope, so the simulated Turkish figures are, if anything, mildly pessimistic. The analytic ceilings use all 8,000 draws and are unaffected. The pooled cross-linguistic design analysis is now running, at per-language samples from 50 up to the 80-participant cap and 100, with thirty simulated studies per condition, alongside a fit of the pooled model to the combined real pilot that will supply its analytic ceiling. Its draw indices are sampled randomly over the full pilot posteriors, so the block limitation above does not apply to it.

Each data-grounded condition rests on 22 to 40 simulated studies, and each literature-anchored cell on 50, so the figures carry a Monte-Carlo error of several percentage points near a power of 80%. Individual cells close to the threshold should be read as near target rather than pinned, and the conclusions rest on the pattern across sample sizes rather than on any single cell. On the literature effects the adequate sample at 72 verbs is reported as at most 50, since the main sweep does not go below 50, and a dedicated low-N arm at the full verb count is estimating it more precisely for that scenario and for Norwegian.

This is an a-priori power analysis, a simulation under assumed effects to size the study, and it is distinct from the confirmatory analysis of the real ratings that will be run once data collection is complete. The analysis code and configuration are openly available, and the report rebuilds from the committed summary tables without the raw data, which remain private to CLAPS until the project reaches a more advanced stage. The report will be finalised once the data-grounded runs complete and the largest English and Turkish samples are in.

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Appendix A
Composition of the Pilot Sample

Appendix A describes the pilot sample from which the affectedness ratings and the descriptive trends in Figure 1 are drawn, and against which the data-grounded analysis fits its models. The counts are read directly from the harmonised pilot data.

Table 3

Composition of the Harmonised Pilot Sample, by Language

Language	Participants	Verbs	Sentence types	Items	Judgements
English	70	72	3	144	30,240
Turkish	70	72	3	144	30,240
Norwegian	57	66	2	132	15,048

Note. Counts are read from the harmonised pilot data. Items are verb-by-agent-gender combinations, so the item count is twice the verb count. Norwegian lacks a pseudo-passive, hence two sentence types rather than three.

Appendix B
Key Code for the Simulation and the Models

Appendix B reproduces the central code, lightly abridged, from the analysis scripts. The code shown is the data-generating process for the literature-anchored cross-check, in `R/06_simulate_design.R`, together with the analysis model in `R/04_model_formulas.R` and the Bayes factor in `R/05_hypothesis_tests.R`. The data-grounded analysis uses the same analysis model and Bayes factor, but draws its data-generating effects, random-effect covariance and thresholds from the fitted pilot models, and reads each verb's affectedness from the pilot data. Its assurance and safeguard variants are in `R/10_simulate_from_pilot.R` and `R/10_simulate_from_pilot_v2.R`.

Each simulated dataset is drawn from the same cumulative-logit model that is later fitted. In the cross-check, affectedness is drawn once per verb across a plausible range, which makes it a verb-level predictor, and the ordinal response follows from the cumulative-logit linear predictor.

```
# DGP for the literature-anchored cross-check, abridged from R/06_simulate_design.R.
verb_semantics <- runif(n_verbs, min = -0.5, max = 0.5)   # one value per verb

# For participant pid, verb vid and sentence type st:
sem   <- verb_semantics[[vid]]
b_int <- if (st == "Active")                beta_active_interaction * sem
         else if (st == "Pseudo_Passive") beta_pseudo_interaction * sem
         else 0
eta   <- part_intercepts[pid] + part_slopes[pid] * sem +
         verb_intercepts[vid] + beta_semantics * sem + b_int

cum_probs <- plogis(thresholds - eta)        # seven ordered categories
probs     <- diff(c(0, cum_probs, 1))
resp      <- sample(seq_len(7), size = 1, prob = probs)
```

The analysis model is the maximal random-effects structure consistent with the design, fitted with the cumulative-logit family.

```
# Analysis model, from R/04_model_formulas.R.
brms::bf(
  Response ~ S_Type * Semantics_scaled +
    (1 + S_Type * Semantics_scaled | Participant) +
    (1 + S_Type | Verb),
  family = brms::cumulative(link = "logit", threshold = "flexible")
)
```

Evidence for each prediction is a directional Savage-Dickey Bayes factor, with the prior sampled alongside the posterior and the posterior mass taken in the predicted direction.

```r
# Directional Savage-Dickey Bayes factor, from R/05_hypothesis_tests.R.
# post_vals and prior_vals are draws of the focal coefficient; the negative
# direction is shown, as for the primary prediction H1b.
post_prob  <- mean(post_vals  < 0)
prior_prob <- mean(prior_vals < 0)
bf_10      <- (post_prob / (1 - post_prob)) / (prior_prob / (1 - prior_prob))
```

Appendix C

Full Bayes-Factor Operating Characteristics

Appendix C reports the operating characteristics for all four predictions in every cell of the corrected run with at least ten replicates, namely the probability of exceeding the primary and secondary Bayes-factor thresholds, the median Bayes factor and the convergence rate.

Table 4

Probability of Exceeding the Primary and Secondary Bayes-Factor Thresholds, Median Bayes Factor and Convergence Rate, by Design Condition and Prediction

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
English	40	50	H1a	74%	96%	2.8e+01	100%	50
English	40	50	H1b	76%	88%	2.1e+01	100%	50
English	40	50	H2a	46%	74%	7.7e+00	100%	50
English	40	50	H2b	0%	0%	1.3e-01	100%	50
English	40	60	H1a	94%	100%	5.3e+01	100%	50
English	40	60	H1b	80%	96%	3.7e+01	100%	50
English	40	60	H2a	68%	94%	2.5e+01	100%	50
English	40	60	H2b	0%	0%	4.1e-02	100%	50
English	40	70	H1a	76%	98%	3.1e+01	98%	50
English	40	70	H1b	86%	96%	8.0e+01	98%	50
English	40	70	H2a	60%	86%	1.6e+01	98%	50
English	40	70	H2b	0%	2%	6.3e-02	98%	50
English	40	80	H1a	80%	98%	2.6e+01	98%	50
English	40	80	H1b	96%	98%	1.8e+02	98%	50
English	40	80	H2a	52%	88%	1.1e+01	98%	50
English	40	80	H2b	0%	0%	9.3e-02	98%	50
English	40	90	H1a	88%	100%	4.9e+01	98%	50
English	40	90	H1b	96%	100%	5.1e+02	98%	50
English	40	90	H2a	58%	88%	1.2e+01	98%	50
English	40	90	H2b	0%	0%	8.1e-02	98%	50
English	72	50	H1a	98%	100%	7.6e+02	92%	50
English	72	50	H1b	98%	100%	8.6e+02	92%	50
English	72	50	H2a	58%	84%	1.5e+01	92%	50
English	72	50	H2b	0%	0%	6.7e-02	92%	50
English	72	60	H1a	100%	100%	1.6e+03	90%	50
English	72	60	H1b	98%	100%	1.2e+03	90%	50
English	72	60	H2a	72%	92%	1.9e+01	90%	50
English	72	60	H2b	0%	2%	5.2e-02	90%	50
English	72	70	H1a	98%	100%	5.8e+02	84%	50
English	72	70	H1b	98%	100%	5.9e+03	84%	50
English	72	70	H2a	60%	86%	1.5e+01	84%	50
English	72	70	H2b	0%	0%	6.6e-02	84%	50
English	72	80	H1a	96%	100%	7.1e+02	82%	50
English	72	80	H1b	100%	100%	7.8e+03	82%	50
English	72	80	H2a	78%	94%	5.5e+01	82%	50
English	72	80	H2b	0%	2%	1.8e-02	82%	50
English	72	90	H1a	100%	100%	1.8e+03	82%	50

(continued)

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
English	72	90	H1b	100%	100%	9.8e+05	82%	50
English	72	90	H2a	80%	88%	5.8e+01	82%	50
English	72	90	H2b	0%	0%	1.7e-02	82%	50
Turkish	40	50	H1a	70%	100%	1.3e+02	100%	10
Turkish	40	50	H1b	80%	100%	2.1e+02	100%	10
Turkish	40	50	H2a	10%	60%	3.5e+00	100%	10
Turkish	40	50	H2b	0%	0%	2.9e-01	100%	10
Turkish	40	60	H1a	74%	90%	7.3e+01	100%	50
Turkish	40	60	H1b	86%	96%	7.9e+01	100%	50
Turkish	40	60	H2a	54%	74%	1.1e+01	100%	50
Turkish	40	60	H2b	2%	2%	8.9e-02	100%	50
Turkish	72	50	H1a	96%	98%	1.8e+03	90%	50
Turkish	72	50	H1b	96%	100%	8.8e+02	90%	50
Turkish	72	50	H2a	58%	86%	1.6e+01	90%	50
Turkish	72	50	H2b	0%	0%	6.4e-02	90%	50
Turkish	72	60	H1a	100%	100%	4.8e+02	94%	50
Turkish	72	60	H1b	96%	100%	1.0e+03	94%	50
Turkish	72	60	H2a	62%	92%	2.1e+01	94%	50
Turkish	72	60	H2b	0%	0%	4.9e-02	94%	50

Note. The two probability columns give the proportion of the simulated studies in which the Bayes factor reached the primary threshold of 10 and the secondary threshold of 3. Median BF is the median Bayes factor across those studies, to two significant figures. Convergence is the proportion of fits meeting the convergence criteria, and Reps is the number of simulated studies in the cell. Cells with fewer than ten replicates are omitted as still accumulating.